

ECON 164: Theory of Economic Growth

Week 9A: Natural Resources and the Environment

Levi Crews

Spring 2026

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 - ...endogenous technological change A wins the race against declining X/L
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Introduction

- **last time:** added factor in **fixed supply** (land X) $\rightarrow$ declining GDPpc unless...
 - ...endogenous technological change A wins the race against declining X/L
 - but we treated land as a **renewable resource**: can use the same land over and over
- **today:** **nonrenewable resources** (petroleum, coal, natural gas, minerals, ...)
 - like last time: race between A vs. X/L and E/L
 - quantify the **drag on growth** from **resource extraction**
 - consider relationship between **GDP per capita** and **pollution**

Production with nonrenewables

Production now also requires **land** X and **energy** E :

$$Y_t = K_t^\alpha E_t^\beta X^\gamma (A_t L_t)^{1-\alpha-\beta-\gamma}$$

notice: **constant returns to scale** in $\{K, E, X, L\}$

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Capital, labor, and technology growth as in the **Solow** model:

$$\dot{K}_t = s_I Y_t - \delta K_t$$

$$\dot{L}_t = g_L L_t$$

$$\dot{A}_t = g_A A_t$$

Resource extraction

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$$g_R = -s_E \rightarrow g_E = -s_E$$

because $E_t = s_E R_t$

How do we solve this model?

Recall the production function:

$$Y_t = K_t^\alpha E_t^\beta X^\gamma (A_t L_t)^{1-\alpha-\beta-\gamma}$$

If we could instead write...

$$Y_t = K_t^\alpha (B_t L_t)^{1-\alpha}$$

...would we know what to do?

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...would we know what to do?

Yes! This is just the **Solow model** with **labor-augmenting productivity** B_t ...

- steady state where $g_{\tilde{k}} \equiv g_K - g_B - g_L = 0 \rightarrow$ **phase diagram**

$$\tilde{k}^{ss} = \left(\frac{s_I}{g_B + g_L + \delta} \right)^{\frac{1}{1-\alpha}}$$

- slope of BGP: $g_y^{ss} = g_B$

What is B_t ?

Taking ratios of the two possible forms of the production function...

$$1 = \frac{K_t^\alpha E_t^\beta X^\gamma (A_t L_t)^{1-\alpha-\beta-\gamma}}{K_t^\alpha (B_t L_t)^{1-\alpha}} = \frac{E_t^\beta X^\gamma A_t^{1-\alpha-\beta-\gamma} L_t^{-(\beta+\gamma)}}{B_t^{1-\alpha}}$$

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...so we can define B_t as

$$B_t \equiv A_t^{\frac{1-\alpha-\beta-\gamma}{1-\alpha}} \left(\frac{E_t}{L_t} \right)^{\frac{\beta}{1-\alpha}} \left(\frac{X}{L_t} \right)^{\frac{\gamma}{1-\alpha}}$$

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...and its growth rate is then

$$g_B = \frac{1-\alpha-\beta-\gamma}{1-\alpha} g_A + \frac{\beta}{1-\alpha} (g_E - g_L) + \frac{\gamma}{1-\alpha} (g_X - g_L)$$

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$$\begin{aligned} g_B &= \frac{1-\alpha-\beta-\gamma}{1-\alpha} g_A + \frac{\beta}{1-\alpha} (g_E - g_L) + \frac{\gamma}{1-\alpha} (g_X - g_L) \\ &= \left(1 - \frac{\beta+\gamma}{1-\alpha} \right) g_A - \frac{\beta}{1-\alpha} s_E - \frac{\beta+\gamma}{1-\alpha} g_L \end{aligned}$$

Defining the resource drag on growth

We can rearrange the expression for the growth rate of B_t to get...

$$g_y^{ss} = g_B = \left(1 - \frac{\beta + \gamma}{1 - \alpha}\right) g_A - \underbrace{\frac{\beta}{1 - \alpha} (sE + g_L)}_{\text{drag from energy } E} - \underbrace{\frac{\gamma}{1 - \alpha} g_L}_{\text{drag from land } X}$$

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Recall: We saw this kind of resource drag at the end of the Malthusian model, too

$$g_y^{\text{ss}} = \beta(g_A^{\text{ss}} - g_L^{\text{ss}}) = (1 - \beta) \frac{\lambda}{1 - \phi} g_L^{\text{ss}} - \underbrace{\beta g_L^{\text{ss}}}_{\text{drag from land } X}$$

- $\alpha \equiv$ share of total factor payments to **capital** $K \approx 0.2$
- $\beta \equiv$ share of total factor payments to **energy** $E \approx 0.1$
- $\gamma \equiv$ share of total factor payments to **land** $X \approx 0.1$

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→ **1.8% observed growth** includes **0.3% drag** from **energy** and **land**

Comparative statics with respect to s_E

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Say we permanently decrease s_E at time $T \rightarrow$ simul. permanent $g_B \uparrow +$ temp. $B_t \downarrow$

1. What happens to $\tilde{k}^{ss}$? (draw the phase diagram!)
2. What happens to g_y^{ss} at T ? In the long run?
3. What happens to $\ln y_t^{\text{BGP}}$ at T ? In the long run?

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- **what's the optimal s_E ?** think about analogy w/ s_I and s_R :
 - s_I in **NGM**: more consumption today vs. more consumption tomorrow
 - s_R in **(semi-)endogenous**: more output today vs. more/better ideas for tomorrow

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- **but something's different**: s_E , unlike s_I and s_R , *does* affect the growth rate g_y^{ss}
- **why?** depletion of R is a linear differential equation $\rightarrow$ like **AK models**

How to value “energy today” vs. “resources for tomorrow”

Suppose the representative household has preferences

$$U_t = (c_t - \bar{c})^{1-\rho} R_t^\rho$$

and faces the *implicit* budget constraint

$$y_t = c_t + p_{Rt} R_t$$

- $\bar{c}$ is **subsistence consumption**, like in the Malthusian model
- household cares directly about the **stock of resources** R_t
 - if **resource extraction** implies **pollution** $\rightarrow$ captures the value of clean environment
 - can reinterpret E_t as **emissions**
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Solving the household's problem

Compute the marginal utility of consumption. . .

$$\text{MU}_c = \frac{(1 - \rho)U_t}{c_t - \bar{c}} = (1 - \rho) \left(\frac{R_t}{c_t - \bar{c}} \right)^\rho$$

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Production and prices

To simplify, let's set $\alpha = \gamma = 0$ to eliminate **capital** (no saving) and **land**...

$$Y_t = E_t^\beta (A_t L_t)^{1-\beta} \quad \rightarrow \quad y_t = A_t \left(\frac{E_t}{A_t L_t} \right)^\beta$$

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... but keep the same resource extraction rule:

$$\dot{R}_t = -E_t$$

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Note the tradeoff: more **energy** $\rightarrow$ more **output** y_t but lower **resources** R_t

$$p_{Rt} = \frac{\partial y_t}{\partial E_t}$$

The optimal choice of s_E

Take the household's optimality condition and the price of resources. . .

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Take the household's optimality condition and the price of resources...

$$\frac{\rho}{1 - \rho} \left(\frac{c_t - \bar{c}}{R_t} \right) = \beta \frac{y_t}{E_t}$$

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Take the household's optimality condition and the price of resources...

$$\frac{\rho}{1-\rho} \left(\frac{c_t - \bar{c}}{R_t} \right) = \beta \frac{y_t}{E_t}$$

...then rearrange to get the optimal choice of s_E :

$$s_{Et} \equiv \frac{E_t}{R_t} = \beta \left(\frac{1-\rho}{\rho} \right) \frac{y_t}{y_t - \bar{c}}$$

Economic growth leads to better environmental quality

$$s_{Et} = \beta \left(\frac{1 - \rho}{\rho} \right) \frac{y_t}{y_t - \bar{c}}$$

- when y_t is close to subsistence level $\bar{c}$...
 - $y_t/(y_t - \bar{c})$ is high
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- requires $\bar{c} > 0$ (nonhomothetic preferences)
 - s_E is constant if $\bar{c} = 0 \rightarrow$ what we assumed to start today

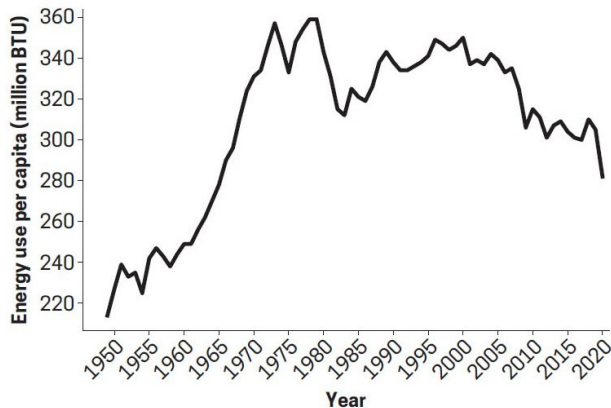
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- requires $\bar{c} > 0$ (nonhomothetic preferences)
 - s_E is constant if $\bar{c} = 0 \rightarrow$ what we assumed to start today
- better model would predict “environmental Kuznets curve” (Brock and Taylor, 2010)
 - s_{Et} rises with y_t at low levels, then falls at high levels

Does economic growth lead to lower resource extraction?

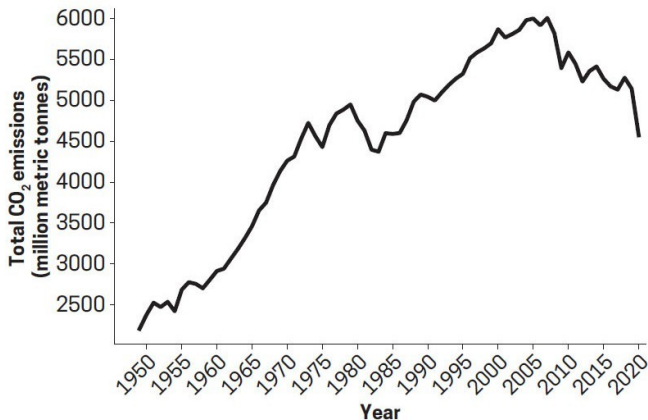
Figure 10.5 Energy Use per Capita in the United States, 1950-2019



SOURCE: Authors' calculations from U.S. Energy Information Administration (2021a).

Does economic growth lead to lower pollution?

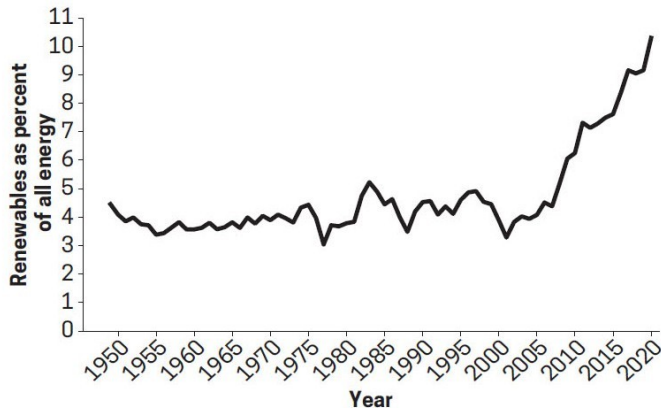
Figure 10.6 Total CO₂ Emissions in the United States, 1950-2019



SOURCE: Authors' calculations from U.S. Energy Information Administration (2021a).

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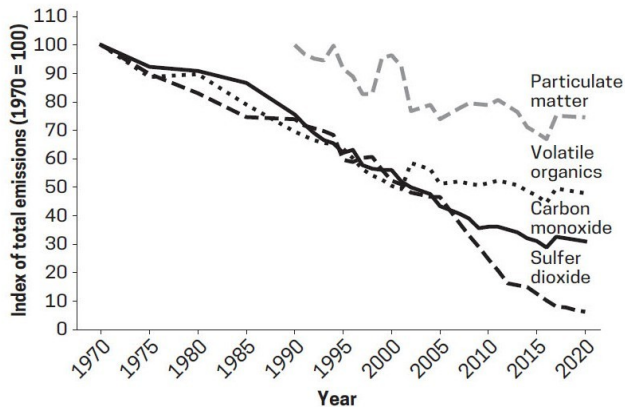
Figure 10.7 Renewables as a Percent of Energy in the United States, 1950-2019



SOURCE: Authors' calculations from U.S. Energy Information Administration (2021a).

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Figure 10.8 Air Pollutants in the United States, 1970–2019



SOURCE: Authors' calculations from U.S. Environmental Protection Agency (2021).

NOTE: The indices for particulate matter are indexed to 100 in 1990, given a lack of prior data. All other series are indexed to 100 in 1970.

Summary

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 - ...endogenous technological change **A** wins the race against declining **X/L**
 - but we treated land as a **renewable resource**: can use the same land over and over

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- last time: added factor in **fixed supply** (land X) $\rightarrow$ declining GDPpc unless...
 - ...endogenous technological change A wins the race against declining X/L
 - but we treated land as a **renewable resource**: can use the same land over and over
- today: **nonrenewable resources** (petroleum, coal, natural gas, minerals, ...)
 - like last time: race between A vs. X/L and E/L
 - can analyze like **Solow model** with **labor-augmenting productivity** B_t
 - find **0.3% drag on growth** from **land** and **resource extraction**
 - some evidence that higher **GDP per capita** leads to lower **emissions** s_E

References

- Brock, William A. and M. Scott Taylor. 2010. "The green Solow model." *Journal of Economic Growth* 15 (2):127–153.
- Dasgupta, Partha and Geoffrey Heal. 1974. "The optimal depletion of exhaustible resources." *Review of Economic Studies* 41 (5):3–28.
- Nordhaus, William D. 1992. "Lethal Model 2: The limits to growth revisited." *Brookings Papers on Economic Activity* 1992 (2):1.